PHYTOTHERAPY RESEARCH
Phytother. Res. **28**: 1315–1319 (2014)
Published online 18 February 2014 in Wiley Online Library
(wileyonlinelibrary.com) **DOI**: 10.1002/ptr.5130

Crocetin, a Carotenoid from *Gardenia jasminoides* Ellis, Protects against Hypertension and Cerebral Thrombogenesis in Stroke-prone Spontaneously Hypertensive Rats

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Crocetin is a natural carotenoid dicarboxylic acid that is found in the fruit of *Gardenia jasminoides* Ellis (Cape Jasmine) and in the stamen and pistil of *Crocus sativus* L. (saffron). It is used worldwide as an important spice, food colorant, and herbal medicine. In the current investigation, we have examined the cardiovascular effects of crocetin using stroke-prone spontaneously hypertensive rats (SHRSPs). Male SHRSPs (6 weeks old) were classified into three groups: a control group and two crocetin groups (25 and 50 mg/kg/day). The animals were given crocetin for 3 weeks. Body weights in each group were not significantly different during the treatment period, but the increase in systolic blood pressures observed with age was significantly moderated by crocetin. Thrombogenesis, assessed using a He-Ne laser technique in pial vessels, was significantly decreased. Antioxidant activity, assessed by measuring urinary 8-hydroxy-2′-deoxyguanosine levels, together with urinary nitric oxide (NO) metabolite levels, was increased significantly after treatment. Acetylcholine-induced vasodilation was measured using the aorta and indicated that endothelial function was significantly improved by crocetin. These results strongly suggest that the antihypertensive and antithrombotic effects of crocetin were related to an increase in bioavailable NO, possibly mediated by decreased inactivation of NO by reactive oxygen species.

Keywords: crocetin; thrombosis; nitric oxide; antioxidant; stroke-prone spontaneously hypertensive rats.

INTRODUCTION

Crocetin is a major active ingredient of traditional herbal medicines derived from the fruits of *Gardenia jasminoides* Ellis and saffron (*Crocus sativus* L.) (Giaccio, 2004). It is also called crocetin-1 or α-crocetin, a biological carotenoid that constitutes the yellow pigment widely used as a natural food colorant (Fig. 1). Crocetin possesses important pharmacological properties, including antioxidant (Ghadrdoost *et al*., 2011; Yoshino *et al*., 2011), antiinflammatory (Nam *et al*., 2010), antiatherosclerotic (Zheng *et al*., 2006), and neuroprotective activities (Ochiai *et al*., 2007; Nam *et al*., 2010), and has been shown to prevent high-fat-diet-induced insulin resistance (Sheng *et al*., 2008). These beneficial effects of crocetin appear to be primarily associated with strong free radical scavenging activity.

It is known that the antioxidant activities of some carotenoids mediate a protective role against cardiovascular diseases (Riccioni, 2009; Rodrigo *et al*., 2011). In particular, carotenoids have been reported to moderate endothelial dysfunction, characterized by enhanced inactivation and/or reduced synthesis of nitric oxide (NO), which have been reported to be risk factors for cardiovascular disease (Rush *et al*., 2005). Napoli and Ignarro (2009) described the role of NO in the pathogenesis of cardiovascular diseases, and in this context, it is evident that reactive oxygen species (ROS) reduce the bioavailability of NO. Subsequently, oxidative stress causes endothelial dysfunction and leads to the progression of atherogenesis and thrombosis.

The aim of the present study was to examine the potential role of crocetin in moderating hypertension and endothelial dysfunction in stroke-prone spontaneously hypertensive rats (SHRSPs). We also focused on the protective effects of crocetin on thrombogenesis in cerebral blood vessels in this well-described animal model of cardiovascular dysfunction.

MATERIALS AND METHODS

Animals and diets. Male SHRSPs, 5 weeks old, were obtained from colonies of specific pathogen-free rats maintained by Japan SLC, Inc. (Hamamatsu, Japan). The rats were used after acclimatization for 1 week (6 weeks old). All experiments were conducted in accordance with the ethical principles of animal care of Kobe Gakuin University (approval nos A110602-9 and A120507-24) and the guiding principles for the care

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Received 11 September 2012
Revised 14 January 2014
Accepted 17 January 2014

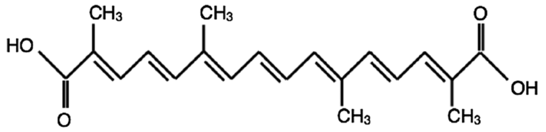

Figure 1. Chemical structure of crocetin.

and use of animals in the field of physiological sciences by The Physiological Society of Japan (The Physiological Society of Japan, 2003).

The standard powder diet for SHRSPs (Funabashi SP diet) was purchased from Funabashi Farm Co., Ltd. (Funabashi, Saitama, Japan). Crocetin (purity; more than 75%) was extracted from *G. jasminoides* Ellis as previously described (Mizuma *et al*., 2009; Kuratsune *et al*., 2010; Umigai *et al*., 2011) and was kindly provided by Dr T. Nakano at Riken Vitamin Co., Ltd. (Tokyo, Japan). In outline, dried gardenia fruits, which contained approximately 1.2% crocin (crocetin glycosides), were treated with hydrous ethanol, and crocetin was extracted by hydrolyzation.

Each animal was kept in an individual cage. The animals were given each powder diet and water *ad libitum*. The weights of the powder diet and water and the body weight of each rat were measured at 9:00 AM every day. Consumption of the diet the day before was assessed from the weight of lost diet of each rat. A control group ($n = 8$) was given an unmodified powder SP diet. In the experiment groups, crocetin was mixed with the powder SP diet to make a dose of 25 mg/kg/day ($n = 8$) or 50 mg/kg/day ($n = 8$), calculated from body weight and consumption of diet of each rat. The animals were fed for 3 weeks from 7 to 10 weeks of age.

Blood pressures were measured by tail cuff method once a week using a LE5001 tail cuff plethysmograph (Panlab, Barcelona, Spain).

Measurement of cerebral thrombogenesis. Closed cranial windows were created (Morii *et al*., 1986), and thrombotic tendency was measured as previously described (Sasaki *et al*., 2011; Ikemura *et al*., 2012). In brief, SHRSPs were anesthetized with sodium pentobarbital (60 mg/kg) and artificially ventilated with a mixture of oxygen in air. A craniotomy was performed using a hand drill to form a cranial window, 5 mm in diameter, in the center of the right parietal bone. A coverslip was placed on the window and fixed with dental resin. Cerebrospinal fluid was continuously infused within the cranial window, and the intracranial pressure was adjusted to 3–5 mmHg to avoid brain herniation. The animals were stabilized in a stereotaxic frame and were placed on the stage of an Olympus BH2 microscope equipped with a long working distance objective lens. Evans blue was administered through a femoral vein. The cerebral vessels were monitored using a charge-coupled device camera (Pulnix, Takenaka System, Kyoto) and recorded on videotape. A He-Ne laser beam (15 μm in diameter) was focused on the center of the selected blood vessels through the optical path of the microscope, and thrombi were formed by repeated irradiation for 10 s at 20-s intervals at a power 13 mW. The number of laser pulses needed to induce the formation of an occlusive thrombus was used as an index of thrombotic tendency.

Determination of urinary 8-hydroxy-2′-deoxyguanosine and nitric oxide metabolites (NO_2/NO_3). Assays were performed before and 3 weeks after each diet. The animals were kept in metabolic cages, and urine samples were collected for 24 h. Samples were stored at −70 °C until assayed. 8-Hydroxy-2′-deoxyguanosine (8-OHdG) was determined using a competitive enzyme-linked immunosorbent assay (ELISA; 8-OHdG check, Japan Institute for the Control of Ageing, Shizuoka, Japan). Urinary nitrite/nitrate (NO_2/NO_3) concentrations, the metabolites of NO, were determined using Griess reagent (Assay kit-C; Dojindo Laboratory, Kumamoto, Japan).

Measurement of vasodilation with acetylcholine. Ring preparations of thoracic aorta were excised after the experiments and dissected in ice cold Krebs-Henseleit solution (composition in mmol/L: 120 NaCl, 4.76 KCl, 25 $NaHCO_3$, 1.18 $NaH_2PO_4 \cdot H_2O$, 1.25 $CaCl_2$, 1.18 $MgSO_4.H_2O$, and 5.5 glucose) to remove connective tissue. Rings (3 mm wide) were cut, and three segments from each animal were mounted on stainless steel hooks and suspended in a water-jacketed 5 mL tissue bath filled with Krebs-Henseleit solution maintained at 37 °C and aerated with 95% O_2 and 5% CO_2 [Easy Magnus System U-5A (four channels), Iwashiya Kishimoto Medical Instruments, Kyoto, Japan]. Aortic rings were stretched to the equivalent of 1.5 g tension for 90 min before commencement of the experiment. After stretching, the aortic rings were equilibrated and contracted with phenylephrine (10^{-5} mol/L). Vascular relaxation was measured by adding acetylcholine at concentrations ranging from 10^{-9} to 10^{-5} mol/L in the animals treated with crocetin (25 mg/kg/day). Relaxation was calculated as a percentage of precontractile vascular tone. The tissue responses were recorded using isometric transducers (Easy Magnus System U-5A) and recorders (Sekonic SS-250F Recorder, Tokyo, Japan).

Statistical analyses. Results are expressed as the number of animals (*N*) and the mean and standard error (SE) of each experiment. A value of $p < 0.05$ was considered to be statistically significant. Statistical evaluation was performed by analysis of variance and by *post hoc* test of Dunnett for body weight and blood pressures, unpaired *t*-test for the changes of 8-OHdG and NO, and Mann–Whitney test for vascular relaxation using a commercially available statistical package (PRISM 5.0; GraphPad Software, Inc., San Diego, CA, USA).

RESULTS

Body weight, systolic blood pressures

Body weight increased over time up to 9 weeks of age. There were no statistically significant differences among the groups as shown in Fig. 2A. The systolic blood pressures of SHRSPs increased with age, and the systolic pressures of control and crocetin rats (25 and 50 mg/kg/day) at 9 weeks old were 239.0 ± 2.6 ($n = 8$), 215 ± 1.3 ($n = 8$; 25 mg/kg/day), and 220 ± 2.8 ($n = 8$; 50 mg/kg/day), respectively. Crocetin administered

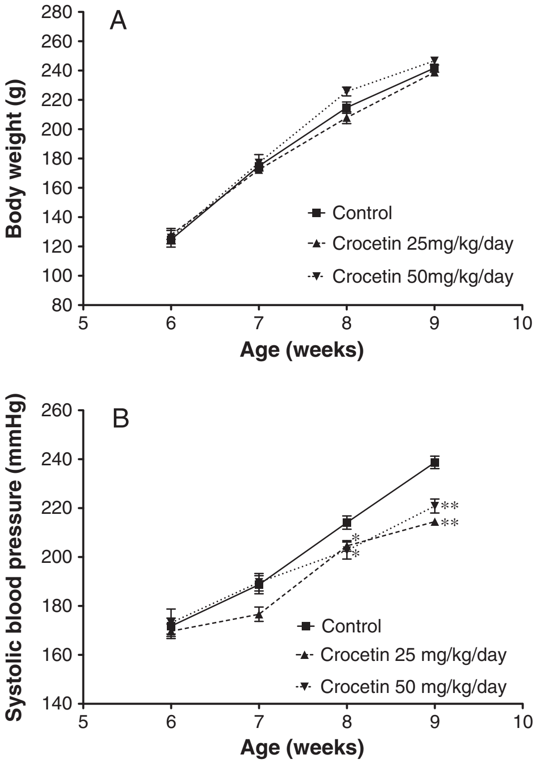

Figure 2. The effects of crocetin on body weight (A) and systolic blood pressure (B) in stroke-prone spontaneously hypertensive rats (SHRSPs). SHRSPs were purchased from a local supplier at 6 weeks old and acclimatized for a week. Crocetin at 25 and 50 mg/kg were fed daily from 7 weeks old. Body weights (A) and systolic blood pressures (B) were measured weekly for 3 weeks. Each bar shows mean ± SE (n = 8). Statistical analysis was performed by one-way analysis of variance and by *post hoc* test of Dunnett. $^*p < 0.05$, $^{**}p < 0.01$ versus control.

at both 25 and 50 mg/kg/day suppressed the increase in systolic blood pressure significantly after treatment for 3 weeks ($p < 0.01$; Fig. 2B).

Effects of crocetin on cerebral thrombosis

Thrombogenesis in cerebral venules for each group of animals, as assessed by the He-Ne laser technique, is summarized in Fig. 3. The number of laser pulses required to generate occlusive thrombi in pial vessels in control and crocetin at 25 and 50 mg/kg/day were 6.8 ± 0.2 (n = 8), 14.1 ± 0.5 (n = 8), and 12.2 ± 0.6 (n = 8), respectively. Thrombus generation was significantly delayed at both concentrations of crocetin ($p < 0.01$).

Effects of crocetin on oxidative stress

The levels of 8-OHdG in urine collected for 24 h before and after dietary intervention, adjusted for body weight, are illustrated in Fig. 4A. There was a significant difference in the amount of urinary 8-OHdG in the control group before and after the 3 weeks experimental period ($p < 0.05$). The amount of urinary 8-OHdG in the crocetin group at 25 mg/kg/day was significantly lower after ingestion than before ($p < 0.05$), and there was a tendency to a decreased level of urinary 8-OHdG before and after ingestion at 50 mg/kg/day ($p = 0.058$).

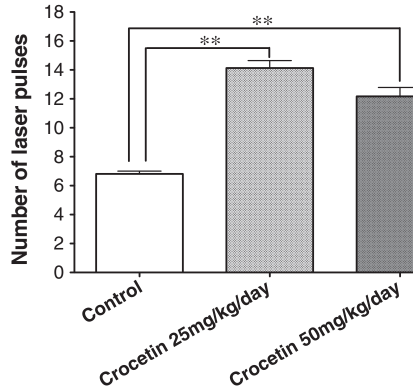

Figure 3. The effects of crocetin on cerebral thrombogenesis in stroke-prone spontaneously hypertensive rats. Cerebral thrombogenesis was assessed after ingestion of crocetin at 25 and 50 mg/kg daily for 3 weeks using a He-Ne laser technique. The number of laser pulses needed to induce the formation of an occlusive thrombus was increased after treatment of crocetin, that is, crocetin inhibited thrombus formation. Each bar shows mean ± SE (n = 8). Statistical analysis was performed by one-way analysis of variance and by *post hoc* test of Dunnett. $^{**}p < 0.01$ versus control. There was no significant difference between crocetin at 25 and 50 mg/kg/day.

Effects of crocetin on nitric oxide metabolites (NO_2/NO_3)

Changes in NO metabolites before and after ingestion of crocetin are illustrated in Fig. 4B. The NO metabolites (NO_2/NO_3) were significantly reduced in the control group after 3 weeks ($p < 0.05$). In contrast, ingestion of crocetin at 25 and 50 mg/kg/day significantly increased NO metabolites (NO_2/NO_3) ($p < 0.05$).

Effects of crocetin on vascular relaxation

Effects of crocetin at 25 mg/kg/day on vascular responses using various concentrations of acetylcholine are shown in Fig. 5. Ingestion of crocetin for 3 weeks increased vascular responses. The percentage relaxation was increased significantly in the treated animals compared with control at acetylcholine concentrations over 10^{-7} mol/L ($p < 0.05$ and $p < 0.01$).

DISCUSSION

Stroke-prone spontaneously hypertensive rats are reported to be exposed to oxidative stress, causing hypertension (Negishi *et al.*, 1999; Mizutani *et al.*, 2001) and enhanced thrombogenesis (Noguchi *et al.*, 1997) in this animal model.

Our present findings demonstrated that crocetin given to SHRSPs for 3 weeks moderated oxidative stress as shown by a significant decrease in urinary 8-OHdG levels. These changes in urinary 8-OHdG levels reflected overall antioxidative capacity in the body and suggested, therefore, that crocetin was a potent antioxidant in these experiments. These antioxidative effects of crocetin have also been examined recently using electron spin resonance signals in the brain of SHRSPs, and the findings confirmed that crocetin could restrict the effects of ROS-related mechanisms in brain diseases such as stroke (Yoshino *et al.*, 2011).

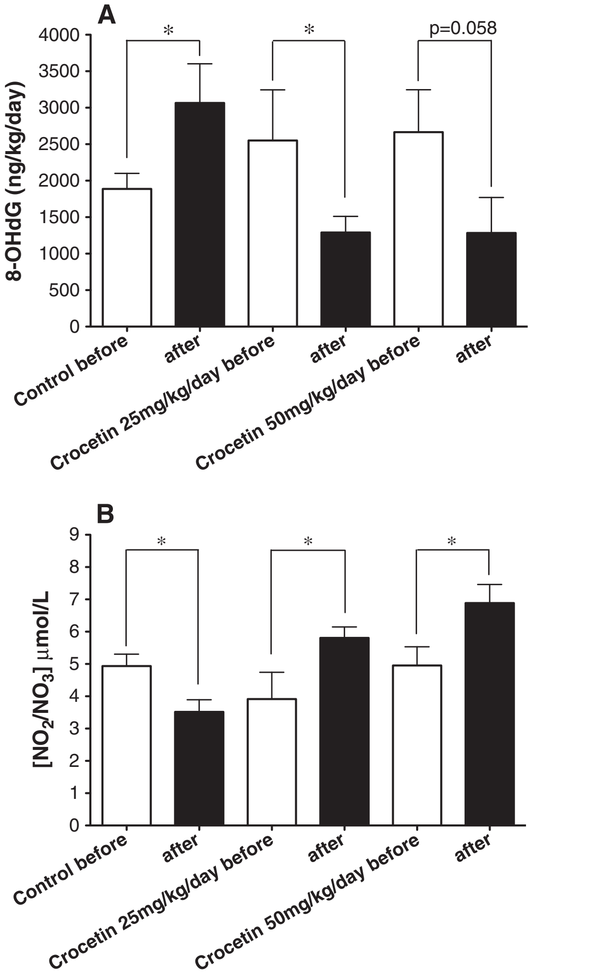

Figure 4. The antioxidant effect (A) and the changes in nitric oxide (NO) metabolites (B) before and after administration of crocetin for 3 weeks in stroke-prone spontaneously hypertensive rats (SHRSPs). (A) The antioxidant activities of crocetin of each SHRSP were assessed by measurement of 8-hydroxy-2′-deoxyguanosine (8-OHdG) in urine samples collected for 24 h in a metabolic cage before and after administration of crocetin for 3 weeks. (B) NO metabolites were measured similarly using the urine samples with Griess reagents. Each bar shows mean ± SE (n = 8). Statistical analysis was performed by unpaired t-test. $^*p < 0.05$.

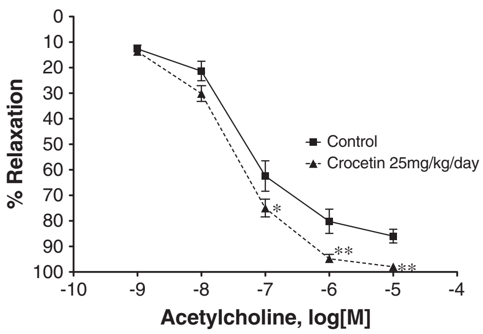

Figure 5. Effect of crocetin on endothelial function. The endothelial function was measured by vasodilation of aortic segments with acetylcholine as described in the Materials and Methods Section. Each bar shows mean ± SE (n = 8). Statistical analysis was performed by Mann–Whitney test. $^*p < 0.05$, $^{**}p < 0.01$ versus control.

In the present study, crocetin inhibited the age-related increase in systolic blood pressures of SHRSPs significantly at 25 and 50 mg/kg/day. This inhibition of increased blood pressure might have been related to the antioxidative effects of crocetin, because SHRSPs are highly exposed to ROS, which scavenges NO and leads to hypertension (Negishi *et al*., 1999; Mizutani *et al*., 2001). As shown in Fig. 2B, the sharp increase in blood pressures with age might underlie the pathogenesis of stroke in SHRSPs, and the increase in NO levels mediated by crocetin in the present study might moderate hypertension and the risk of stroke. We measured the NO derivatives nitrite/nitrate using Griess reagent rather than NO itself, and it is likely that the enhanced levels of NO in SHRSPs were produced not only by endothelial NO synthase (eNOS) but also by inducible NO synthase (iNOS). Yan *et al*. and Hong and Yang reported that crocetin inhibited both lipopolysaccharide-induced iNOS activation and iNOS activity in the course of inflammation and during hemorrhage shock (Yang *et al*., 2006; Yan *et al*., 2010). They concluded that crocetin blocked inflammatory cascades by inhibiting ROS production and iNOS activities. In addition, Tang *et al*. found that crocetin increased NO levels in hypercholesterolemic rabbits by enhancing eNOS activity without affecting iNOS activity (Tang *et al*., 2006). These findings suggested that crocetin significantly restored endothelial function, may be by increasing blood-vessel eNOS activity, leading to elevation of NO production. In a previous study, we measured thrombotic tendency in SHRSPs after the administration of imidapril, an angiotensin-converting enzyme inhibitor known to increase NO. The administration of imidapril and N^G-nitro-L-arginine methyl ester (L-NAME), an inhibitor of NOS, reversed NO levels and moderated the antihypertensive and antithrombotic effects of imidapril (Sasaki *et al*., 2000). In addition, the animals administered L-NAME alone became severely hypertensive, and some of the rats died from stroke. In the current experiments, it appears that crocetin regulated NO synthesis in relation to the pathophysiological ROS levels, and it is conceivable that L-NAME could similarly reverse these effects of crocetin. Negishi *et al*. reported that 24-h urinary 8-OHdG levels were significantly higher in SHRSPs than in age-matched normotensive Wistar–Kyoto rats (Negishi *et al*., 2000). It might be, therefore, that crocetin-administered normotensive control rats did not mediate changes in NO status because ROS production would be expected to be low in these age-matched normotensive animals.

In the present study, we demonstrated potent antithrombotic effects of crocetin using a He-Ne laser technique in the cerebral vessels. Thrombogenesis in this technique mainly reflects platelet responses, and these findings suggested that increased bioactive NO inhibited thrombus formation by governing platelets function, contributing to the beneficial synergistic effects of crocetin. Other investigations have indicated that crocetin possesses a number of important pharmacological properties, including antiatherosclerotic (Zheng *et al*., 2006) and antiplatelet activities (Liakopoulou-Kyriakides and Skubas, 1990; Yang *et al*., 2008). The inhibitory effects of crocetin on platelet activities remain uncertain, however, although Yang *et al*. suggested that the suppressive effects were related to the inhibition of Ca^{2+} mobilization mediated by reductions in both intracellular Ca^{2+} release and extracellular Ca^{2+} influx in platelets (Yang *et al*., 2008).

In conclusion, daily ingestion of crocetin promoted antihypertensive and antithrombotic effects in SHRSPs.

The findings offer the challenging possibility that daily ingestion of crocetin could have important beneficial effects on cardiovascular diseases.

Acknowledgements

The authors are grateful to Dr Takahisa Nakano at Riken Vitamin Co., Ltd. for providing the crocetin. This work was supported in part by the 'Academic Frontier' Project for Private Universities: matching fund subsidy from the Ministry of Education, Science, Sports, and Culture of Japan.

Conflict of Interest

The authors declare that they have no conflict of interest.

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